

Model Development Phase Template

Date	28 June 2025
Student ID	SWUID20250181505
Project Title	online payments fraud detection using machine learning
Maximum Marks	5 Marks

Feature Selection Report Template

In the forthcoming update, each feature will be accompanied by a brief description. Users will indicate whether it's selected or not, providing reasoning for their decision. This process will streamline decision-making and enhance transparency in feature selection.

Feature	Description	Selected (Yes/No)	Reasoning
Step	An integer representing the time step (hour) of the transaction in a simulated 30-day period.	Yes	It captures the temporal context of transactions, which may correlate with fraud patterns (e.g., fraud occurring at specific times). Provides valuable sequential information for the model.
Type	Categorical feature indicating the transaction type (CASH_IN, CASH_OUT,	yes	It is highly relevant to fraud detection, with fraudulent transactions often associated with TRANSFER and CASH_OUT types, as observed in

	DEBIT, PAYMENT, TRANSFER).		bivariate analysis. Encoded using LabelEncoder for model compatibility.
Amount	Float representing the transaction amount in the local currency.	Yes	Selected because fraudulent transactions frequently involve higher amounts, as shown in bivariate analysis. A critical predictor for identifying suspicious transactions.
nameOrig	String identifier for the transaction originator (e.g., customer ID).	No	Excluded because it is a unique identifier with no predictive value for fraud detection. Including it would increase computational overhead without contributing to model performance.
oldbalance Org	Float representing the originator's account balance before the transaction.	Yes	Selected as it provides context for the transaction's legitimacy. Anomalies (e.g., large amount with low oldbalanceOrg) are indicative of fraud, as seen in multivariate analysis.
newbalanc eOrig	Float representing the originator's account balance	Yes	Selected because it complements oldbalanceOrg, showing balance changes. Fraudulent transactions often result in newbalanceOri = 0 for large

	after the transaction.		transfers, a key pattern for detection.
nameDest	String identifier for the transaction recipient (e.g., customer or merchant ID).	No	Excluded as it is a unique identifier with no direct predictive power for fraud detection. Similar to nameOrig, it adds unnecessary complexity to the model.
oldbalanceDest	Float representing the recipient's account balance before the transaction.	Yes	Selected because it provides insight into the recipient's account state, which may reveal fraud patterns (e.g., large transfers to accounts with low balances).
newbalanceDest	Float representing the recipient's account balance after the transaction.	yes	Selected to complement oldbalanceDest, capturing balance updates. Anomalous balance changes in fraudulent transactions enhance model accuracy.
isFraud	Binary label (0 for non-fraudulent, 1	NO	Excluded from input features as it is the target variable for the machine learning model, not a predictor. Used as the

	for fraudulent) indicating whether the transaction is fraudulent.		dependent variable for training and evaluation.
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